

Module - 1

1. Syntax.

Definition - Syntax is the set of rules that tells us how to write a correct C/C++ program. If the syntax is wrong, the programme gives a syntax error.

Basic elements of syntax-

- Header file.
- Namespace
- Main function
- Variables
- Input & output.
- Semicolon.

2. Conditions.

Definition - Conditions are used to make decisions in a program

Types of Conditional statements

- | | |
|-----------------------|----------------------|
| (a) if statement | (d) Switch statement |
| (b) if-else statement | |
| (c) else if ladder | |

3. Loops:-

Definitions - loops are used to repeat a block of code multiple times.

Types of loops

- a) for loop - used when no. of iteration is known.
- b) while loop - checks the condition before executing the loop.
- c) do while loop - Executes the loop at least once.

~~##~~



```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5;
5
6     for(int i = 1; i <= n; i++) {
7         for(int j = 1; j <= n; j++) {
8             if(i == 1 || i == n || j == 1 || j == n)
9                 printf("*");
10            else
11                printf(" ");
12        }
13        printf("\n");
14    }
15
16    return 0;
17 }
```

```

*****
*       *
*       *
*       *
*       *
*****

```



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Main.c

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Console

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AI Agent

238 ms

```
1  #include <stdio.h>
2
3  int main() {
4      int n = 5;
5
6      for(int i = 1; i <= n; i++) {
7          for(int j = 1; j <= i; j++) {
8              printf("*");
9          }
10         printf("\n");
11     }
12
13     return 0;
14 }
```

```
*
**
***
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```

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```
1  #include <stdio.h>
2
3  int main() {
4      int n = 5;
5
6      for(int i = n; i >= 1; i--) {
7          for(int j = 1; j <= i; j++) {
8              printf("*");
9          }
10         printf("\n");
11     }
12
13     return 0;
14 }
```


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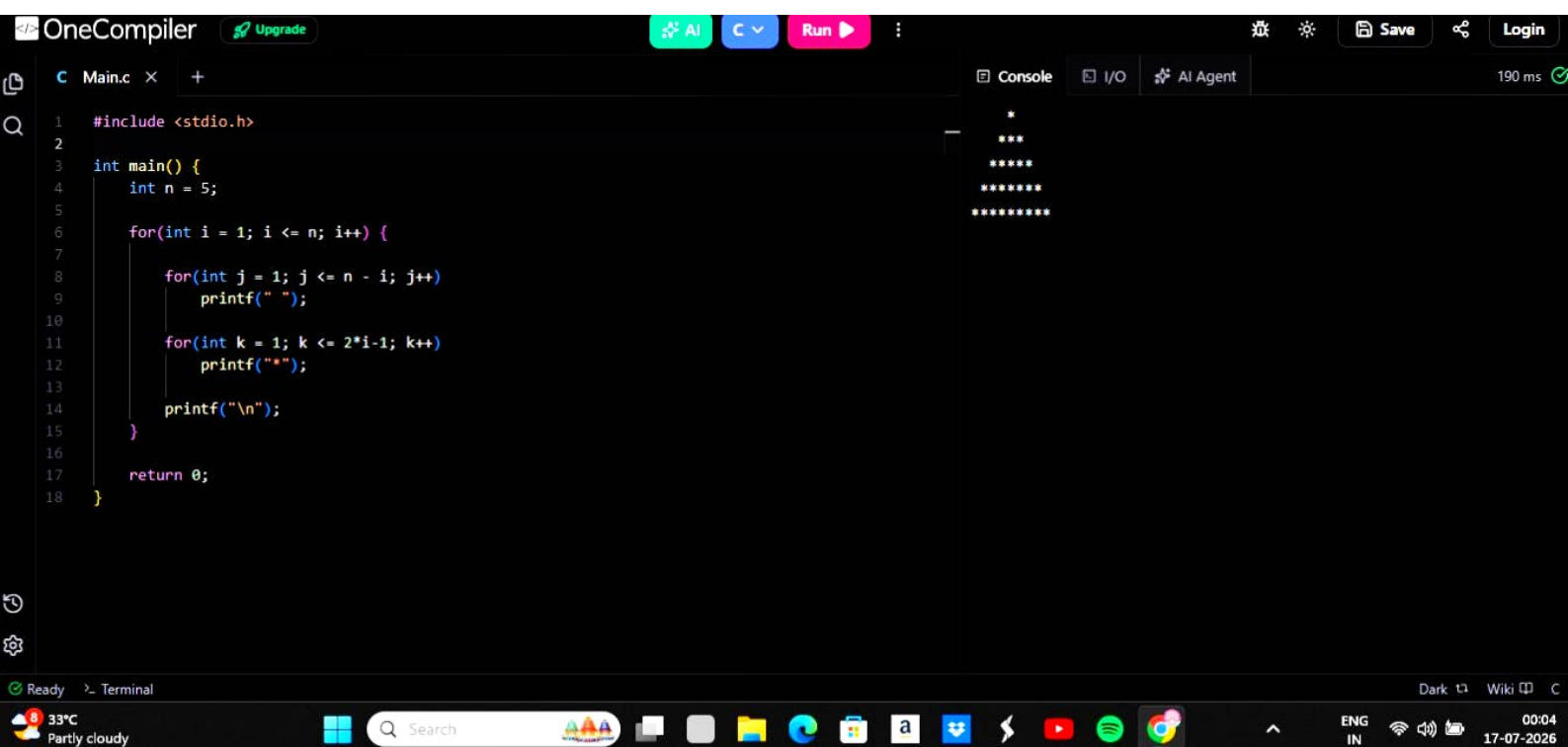
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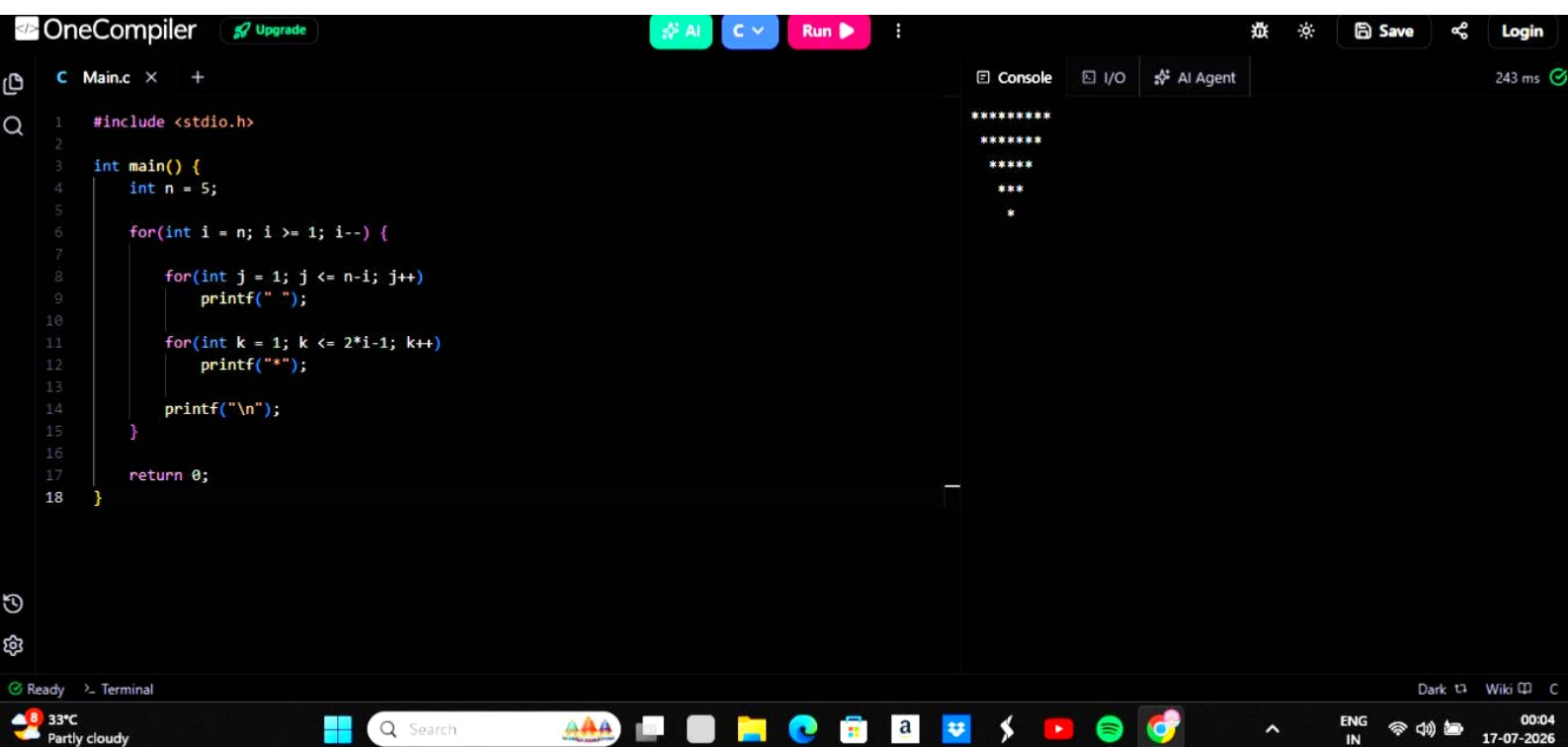
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Main.c

1#include <stdio.h>

2

3int main() {

4 int n = 5;

5

6 // Upper Half

7 for(int i = 1; i <= n; i++) {

8 for(int j = 1; j <= i; j++)

9 printf("*");

10 printf("\n");

11 }

12

13 // Lower Half

14 for(int i = n-1; i >= 1; i--) {

15 for(int j = 1; j <= i; j++)

16 printf("*");

17 printf("\n");

18 }

19

20 return 0;

21 }

Console

I/O

AI Agent

180 ms

*
**

**
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